

# References

Academic research, domain context, and tooling sources — Freight Rate Forecasting & SAIL Coal Import Logistics

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## Academic Research — Freight Rate Forecasting

- Katris, C. (2021). “Time series forecasting methods for the Baltic dry index,” *Journal of Forecasting* — compares ARIMA, FARIMA, neural networks and other methods for forecasting the daily BDI. [Wiley Online Library](#)
- “A Deep Learning Framework for Baltic Dry Index Forecasting,” *ScienceDirect* — proposes deep learning-based feature extraction for BDI prediction, noting BDI's highly non-linear, non-stationary behavior. [ScienceDirect](#)
- Batchelor, Alizadeh & Visvikis (2007), “Forecasting spot and forward prices in the international freight market,” *International Journal of Forecasting*

## Domain Context — SAIL / Coal Import Logistics

- Cabinet approval allowing SAIL and RINL to directly charter vessels for coal imports without going through TRANSCART — background on why chartering decision-making matters for steel PSUs, which import 14–15 million tonnes of coking coal annually. [Business Standard](#)
- CAG report (2025) flagging that SAIL consumed imported coal above management-fixed norms between 2016–2023, incurring ~Rs. 2,539 crore in additional expenditure — a real, cited example of the cost impact of suboptimal import/logistics decisions. [Business Standard](#)
- *Coal Age* — “India Coal Logistics Perspective,” on which East Coast ports (Paradip, Gangavaram) can handle Capesize vs Panamax vessels, and how port infrastructure constraints affect chartering choices. [Coal Age](#)
- SAIL sourcing coking coal shipments from Russia at ~75,000 tonnes/vessel — real example of cargo volumes relevant to vessel-type sizing. [Deccan Herald](#)

## Tools & Documentation

- [Facebook/Meta Prophet](#) — official forecasting library documentation
- Baltic Exchange — source methodology for the Baltic Dry Index

## Data Source Used in Prototype

- Historical Baltic Dry Index dataset (1985–2013), used as a public proxy for freight rate trends in the absence of real-time SAIL-specific chartering data

This mix of peer-reviewed forecasting research, real-world domain grounding — including SAIL's actual chartering context and a cited cost-inefficiency example — and the specific tools used gives the methodology concrete, verifiable grounding.